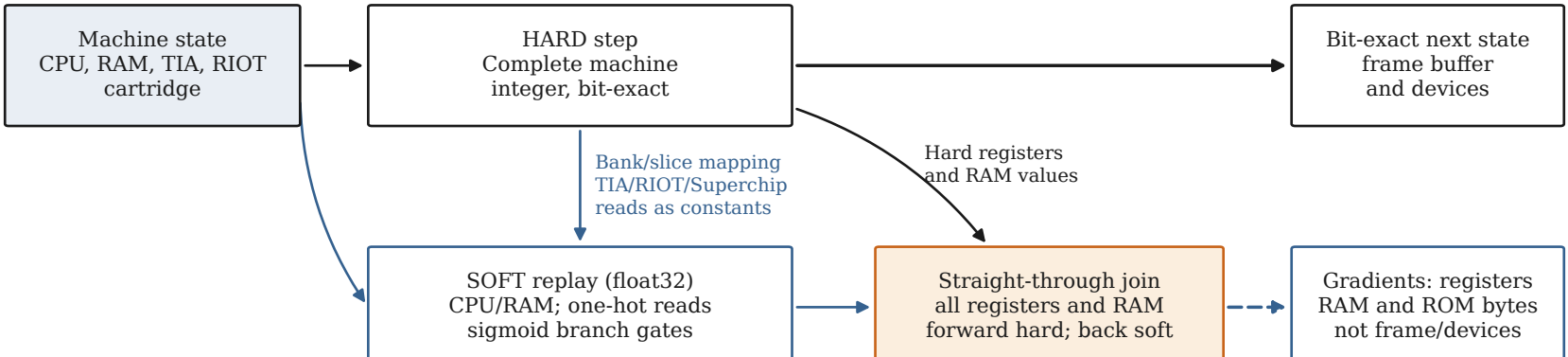
Machine state
CPU, RAM, TIA, RIOT
cartridge
HARD step
Complete machine
integer, bit-exact
Bit-exact next state
frame buffer
and devices
Bank/slice mapping
TIA/RIOT/Superchip
reads as constants
Hard registers
and RAM values
SOFT replay (float32)
CPU/RAM; one-hot reads
sigmoid branch gates
Straight-through join
all registers and RAM
forward hard; back soft
Gradients: registers
RAM and ROM bytes
not frame/devices